

ECO101H1F L201 & L301: Principles of Microeconomics
Midterm 1 (Version A), Fall 2023

University of Toronto, Faculty of Arts and Science
Instructor: Kripa Freitas

Multiple Choice Questions

PUT MULTIPLE CHOICE ANSWERS IN THE BUBBLE SHEET (LAST PAGE).

No work here will be considered.

- 1) [1 point] Colour in any letter on the bubble sheet to receive points for this question. This ECO101 midterm is for...
- A. Prof. Freitas' sections only
 - B. Prof. Freitas' sections only
 - C. Prof. Freitas' sections only
 - D. Prof. Freitas' sections only
 - E. Prof. Freitas' sections only
- 2) [2 points] Mita's total benefit from the number of yoga classes a week is given in the table below. She pays \$25 per class. How many classes does she attend a week?

No. of classes a week	1	2	3	4
Total Benefit	\$40	\$70	\$90	\$100

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

Solution: C. The largest quantity for which $MB \geq MC$ is 2 classes. At that quantity Total Benefit is also at least Total Cost, so she chooses it.

No. of classes a week	1	2	3	4
Total Benefit	\$40	\$70	\$90	\$100
Marginal Benefit	\$40	\$30	\$20	\$10
Marginal Cost	\$25	\$25	\$25	\$25
Total Cost	\$25	\$50	\$75	\$100
$TB - TC$	\$15	\$20	\$15	\$0

- 3) [2 points] Haidar bought a ticket to the TIFF festival for \$25. He has to use it to watch one of the three movies (with his valuation for each in \$) in the table below. Based on this information, what is his Opportunity Cost of using the ticket to watch *Hate to Love: Nickelback*?

Movie	Valuation in \$s
The Boy and the Heron	\$50
Dear Jassi	\$30
Hate to Love: Nickelback	\$20

- A. \$80
- B. \$75
- C. \$55
- D. \$50
- E. \$30

Solution: D. The \$25 is sunk, which means the only resource he uses is the ticket itself. Spending the ticket on *Nickelback* means giving up its next best use, watching *The Boy and the Heron*, which he values at \$50. His Opportunity Cost is therefore \$50.

- 4) [2 points] Clothilde and Liao split their time between baguette and bun production. The amount of each they can produce in one hour is given in the table below. Each has 10 hours of time. Based on this information, what is Clothilde's Opportunity Cost of an extra bun?

Person	Baguettes	Buns
Clothilde	15 per hour	25 per hour
Liao	20 per hour	30 per hour

- A. $\frac{3}{5} = 0.6$ baguettes
- B. $\frac{2}{3} = 0.67$ baguettes
- C. $\frac{5}{3} = 1.67$ baguettes
- D. $\frac{6}{5} = 1.2$ baguettes
- E. $\frac{3}{2} = 1.5$ baguettes

Solution: A. With one hour she can produce 15 baguettes or 25 buns. An extra bun therefore takes her $\frac{1}{25}$ of an hour, and in that time she could have produced

$$\frac{15}{25} = \frac{3}{5} = 0.6 \text{ baguettes.}$$

- 5) [2 points] Jie and Rahman split their time between cleaning yards and trimming trees. The time they need to do one unit of each is given in the table below. Each has 10 hours of time. If both spend all their time cleaning yards, the maximum number of clean yards they could produce is...

Person	per yard cleaned	per tree trimmed
Jie	1 hour	3 hours
Rahman	2 hours	4 hours

- A. 5.83
- B. 15
- C. 30
- D. 40
- E. 70

Solution: B. In 10 hours Jie can clean 10 yards, while Rahman can clean $\frac{10}{2} = 5$. In total that gives us 15.

- 6) [2 points] The perfectly competitive market for cups of coffee has downward-sloping demand and upward-sloping supply. The price of coffee beans increases. As a result, our prediction is market price of coffee cups will ____ and market quantity will ____.

- A. decrease; decrease

- B. increase; decrease
- C. decrease; increase
- D. increase; could increase or decrease
- E. could increase or decrease; decrease

Solution: B. Dearer beans raise MC, which is a shift up (in) of the supply curve. That moves us along the demand curve to a higher price and a lower quantity.

- 7) [2 points] Consumers choose between chocolate mousse and vanilla ice-cream for dessert. The perfectly competitive market for vanilla ice-cream has downward-sloping demand and upward-sloping supply. The price of chocolate mousse increases. As a result, our prediction is market price of vanilla ice-cream will ____ and market quantity will ____.
- A. increase; increase
 - B. increase; decrease
 - C. decrease; increase
 - D. decrease; decrease
 - E. could increase or decrease; decrease

Solution: A. An increase in the price of a substitute shifts the demand curve out, which moves us along the supply curve to a higher price and a higher quantity.

- 8) [2 points] Quantity demanded (Q) for pens (in millions) at every market price P is given by $Q = 30 - 3P$. At a price of \$6, the (absolute) elasticity of demand for pens is...
- A. 0.67
 - B. 0.167
 - C. 1.5
 - D. 6
 - E. None of the above.

Solution: C. At $P = 6$ we have $Q = 30 - 3(6) = 12$, so

$$|\varepsilon_D| = \left| \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} \right| = \left| -3 \cdot \frac{6}{12} \right| = 1.5$$

- 9) [2 points] Anna's demand for bubble tea is downward sloping. At the current market price of \$7 for bubble tea, her elasticity of demand is 1.3. If the price of bubble tea increases, Anna's spending on bubble tea will...
- A. stay the same.
 - B. could increase or decrease.
 - C. could decrease or stay the same.
 - D. increase.
 - E. decrease.

Solution: E. With elastic demand, when the price increases quantity demanded decreases by a greater percentage, which reduces spending.

- 10) [3 points] Ayaka will get a pet. She is choosing between a dog and a cat. The price of a dog from a breeder is \$2,000 and a cat is \$1,000. Dogs need to be walked twice a day, which takes time; cats do not. The costs of all other spending on the cat and dog (e.g. price of food, vet bills, etc.) are exactly the same. Based only on this information, if the breeder gives Ayaka a discount on the dog making the price \$500 instead, her Opportunity Cost of owning a dog ____ and her Willingness to Pay for a dog ____.

- A. decreases; stays the same.
- B. increases; stays the same.
- C. decreases; decreases.
- D. increases; decreases.
- E. decreases; increases.

Solution: A. She buys the dog if $V_{dog} \geq OC_{dog}$, where the Opportunity Cost of the dog is what she pays for it, the resources it uses up, and the net benefit she gives up by not getting the cat:

$$OC_{dog} = \$price_{dog} + Resources_{dog} + \$V_{cat} - \$price_{cat} - \$Resources_{cat}$$

Her Willingness to Pay for the dog is the highest $\$price_{dog}$ at which that still holds, so setting the two sides equal and solving for the price gives

$$WTP_{dog} = \$V_{dog} - Resources_{dog} - \$V_{cat} + \$price_{cat} + \$Resources_{cat}$$

When $\$price_{dog}$ falls the Opportunity Cost falls with it, but the price of the dog doesn't appear in the WTP expression at all, so WTP stays the same.

- 11) [3 points] Tickets to a comedy show on Dec 12 are priced at \$50, but they are sold out. You manage to get a ticket from a friend for \$70. The final exam schedule is released and you have a final scheduled at the same time as the show, so you cannot go. Charlie is willing to give you \$10 for your ticket and Elliott will give you \$20. If you decide to sell your ticket to Max instead, the lowest price Max would need to pay you is...
- A. \$0.01
 - B. \$10.01
 - C. \$20.01
 - D. \$50.01
 - E. \$70.01

Solution: C. The \$70 you paid is sunk. Without Max, if you don't use the ticket your next best option is to sell it to Elliott for \$20, so that is your Opportunity Cost of selling to Max. Any amount above \$20 beats it.

- 12) [3 points] Kwame allocates his time between designing video games and building gaming computers. He purchases software which decreases the amount of time it takes to design a video game, all else stays the same. As a result, his Opportunity Cost for video games _____ and his Opportunity Cost for gaming computers _____.
- A. increases; stays the same.
 - B. decreases; stays the same.
 - C. increases; decreases.
 - D. decreases; increases.
 - E. stays the same; increases.

Solution: D. Now that he's faster at designing video games, he gives up fewer computer builds to design a game, which decreases his Opportunity Cost of video games. Building one computer still takes the same time, but in that time he could now have produced more games, so his Opportunity Cost of computers increases.

- 13) [3 points] Morticia and Wednesday can print T-shirts or print hoodies using time. Each has 20 hours of time and the number of hours they need to print 1 of each is given in the table below. What trade will make them both better off?

	1 Hoodie printed	1 T-shirt printed
Morticia	2 hours	6 hours
Wednesday	2 hours	4 hours

- A. 1 hoodie for 0.2 T-shirts
- B. 1 hoodie for 0.4 T-shirts
- C. 1 hoodie for 1.5 T-shirts
- D. 1 hoodie for 2.5 T-shirts
- E. 1 hoodie for 3.5 T-shirts

Solution: B. The table below shows the Opportunity Cost of a hoodie for each of them. Morticia gives up less to print a hoodie, so she has the comparative advantage in hoodies: **Morticia** should export hoodies and **Wednesday** should export T-shirts. An exchange rate that makes them both better off lies between 0.33 and 0.5 T-shirts for 1 hoodie.

	Morticia	Wednesday
OC of 1 hoodie	$\frac{2}{6} = 0.33$ T-shirts	$\frac{2}{4} = 0.5$ T-shirts

- 14) [3 points] Consider the perfectly competitive market for polyester sweaters. At the current market price, the (absolute) own-price elasticity of demand is 1.7, the elasticity of supply is 0.6 and the income elasticity is -0.2 . Incomes increase and the price of polyester decreases. Market price ____ and market quantity ____ for polyester sweaters.
- A. decreases; stays the same.
 - B. could increase or decrease; decreases.
 - C. could increase or decrease; increases.
 - D. increases; could increase or decrease.
 - E. decreases; could increase or decrease.

Solution: E. Demand decreases when income rises, as the sweaters are an inferior good (the income elasticity is negative). A decrease in the price of an input lowers MC and so increases supply. Both of these push market price down, but they push quantity in opposite directions, so the net effect on quantity depends on which curve moves relatively more.

- 15) [3 points] Assume a perfectly competitive market for new bikes with downward-sloping demand and upward-sloping supply. At the current market price, the (absolute) own-price elasticity of demand is 0.5 while the elasticity of supply is 1.5. New bike lanes in the city encourage people to bike to work instead of driving, all else stays the same. If this causes the quantity of bikes sold to change by 10%, we would predict a ____ change in market prices.
- A. 2%
 - B. 5%
 - C. 6.67%
 - D. 15%
 - E. 20%

Solution: C. As demand for bikes increases we move along the supply curve, and hence use the elasticity of supply:

$$\varepsilon_S = \frac{\% \Delta Q}{\% \Delta P} \implies \% \Delta P = \frac{\% \Delta Q}{\varepsilon_S} = \frac{10\%}{1.5} = 6.67\%$$

- 16) [3 points] We have two sellers (Anita and Balewa) whose Marginal Costs of production for units of the good are given in the table below. At what market price will we get exactly 4 units supplied?

Quantity	1	2	3	4
Anita	\$19	\$23	\$27	\$29
Balewa	\$15	\$18	\$21	\$26

- A. \$20
- B. \$22
- C. \$24
- D. \$26
- E. \$29

Solution: B. Arranging the Marginal Costs from lowest to highest, as shown below, at a price of \$22 we get 3 units from Balewa and one unit from Anita. These are the only units that pass the $P \geq MC$ threshold.

Q	1	2	3	4	5
MC	\$15	\$18	\$19	\$21	\$23

- 17) [3 points] The perfectly competitive market for carpets has (absolute) own-price elasticity of demand = 0.7 and elasticity of supply = 1.6 at the current market price. A change in the market causes quantity sold in the market to increase by 10%. Based only on this information, we can predict that consumer spending on carpets...
- A. increases.
 - B. decreases.
 - C. stays the same.
 - D. could stay the same or increase.
 - E. could increase or decrease.

Solution: E. Quantity has increased, but we don't know why. If it is due to an increase in supply, then we move down the demand curve to a lower price and higher quantity, and with inelastic demand that causes spending to decrease. If it is due to an increase in demand, then moving along the supply curve to a higher price and higher quantity would increase spending.

- 18) [3 points] Consider the perfectly competitive market for new phones with downward-sloping demand and upward-sloping supply. Buyers choose between buying new or used phones. Sellers produce either new phones or new TVs with the same resources. The price of a used phone increases and the price of a new TV increases. In the market for new phones, we would expect this to cause market price to ____ and market quantity to ____.
- A. either increase or decrease; increase.
 - B. either increase or decrease; decrease.
 - C. increase; increase.
 - D. decrease; either increase or decrease.
 - E. increase; either increase or decrease.

Solution: E. The price increase of the next best alternative increases the MWTP for new phones, shifting demand out. An increase in the price of a production alternative increases the MC of producing new phones, which shifts supply in. This results in a higher price, but quantity could increase or decrease.

- 19) [3 points] Assume that new houses are a normal good transacted in a perfectly competitive market. For 2026 to 2027, there was no change in the market price of a new home, but the quantity transacted increased. The change from 2026 to 2027 can be explained by ____ in incomes and ____ in the price of lumber used in housing construction.
- A. an increase; a decrease
 - B. an increase; an increase
 - C. a decrease; a decrease
 - D. a decrease; an increase
 - E. an increase; no change

Solution: A. The change in income shifts the demand curve, while the change in the price of lumber shifts the supply curve. To increase the quantity transacted, demand must shift out (incomes must increase, as houses are a normal good) and supply must also increase (a decrease in MC due to a decrease in lumber prices). Each shift on its own would move the price, but they move it in opposite directions: if the two shifts cancel out exactly, we get no change in price alongside an increase in market quantity.

- 20) [3 points] Over the weekend, you could spend your free time either going to the cinema or going for yoga classes. Your Willingness to Pay for an extra movie a month is downward sloping. With your TIFF subscription, for \$80 a month you have access to unlimited movies, and right now you watch 4 movies in the cinema a month. The price of your yoga class increases. When this happens, your Marginal Willingness to Pay for the 4th movie ____ and you would now watch ____ movies a month.
- A. remains the same; exactly 4
 - B. increases; 4 or fewer than 4
 - C. increases; 4 or more than 4
 - D. decreases; 4 or fewer than 4
 - E. decreases; 4 or more than 4

Solution: C. As the value of your next best option gets worse, your MWTP for movies increases. The subscription means the MC of an extra movie is still \$0, so at the same $MC = 0$ you watch more movies in general — or, if you're thinking about movies as discrete, either the same number or more.

PUT MULTIPLE CHOICE ANSWERS IN THE BUBBLE SHEET (LAST PAGE).

No work here will be considered.

Short Answer Questions

- Q1) Rachel works 50 hours a week at the coffee shop, Central Perk. She gets paid \$800 per week and earns additional money from tips. As she uses the TTC to commute everywhere, she buys an annual TTC pass every year on Jan 1st. A bookshop next to the coffee shop offers her \$1,000 per week for 40 hours of work a week. If she accepts she would start Nov 1st.

- Q1a) [5 points] Based on the given information, what is Rachel's Opportunity Cost of accepting the bookshop's job offer? Write down an equation for the Opportunity Cost and briefly explain how you derived it.

Solution: The benefit from working at the bookshop is the wage earned, \$1,000. Against that:

- **The resources used:** 40 hours of time. (The TTC pass is a sunk cost — she buys it on Jan 1st either way.)
- **The next best option** is working at the current job, worth $\$800 + \text{Tips}$ and using 50 hours.

$$OC = 40 \text{ hrs} + \$800 + \text{Tips} - 50 \text{ hrs} = \$800 + \text{Tips} - 10 \text{ hrs}$$

- Q1b)** [5 points] Based only on the information given, Rachel should accept the bookstore's offer. Agree, disagree, or it depends? Explain your reasoning.

Solution: It depends. We can't settle it without knowing how she values her tips and her time.

Rachel will accept the job if the benefit is at least as high as the Opportunity Cost. The benefit is \$1,000:

$$\$1,000 \geq \$800 + \text{Tips} - 10 \text{ hrs} \quad \Longleftrightarrow \quad \$200 + 10 \text{ hrs} \geq \text{Tips}$$

So she should accept only if the \$200 of extra wage plus the value of the 10 hours of freed-up time is worth more than the tips she earned at the coffee shop. We need more information on how she values those 10 hours and on how much she makes in tips.

- Q1c)** [5 points] Rachel has a (very well-behaved) dog. She had to leave the dog at home when working at the coffee shop, which made both Rachel and the dog sad. The coffee shop now allows Rachel to bring her dog into work with her. How does this affect the minimum amount the bookstore needs to pay Rachel to accept the job at the bookstore? Explain your reasoning.

Solution: It increases it. Rachel makes her decision by comparing benefits against Opportunity Cost, and her Opportunity Cost of accepting the bookstore offer is now higher, because she is giving up a more valuable job.

$$\$wage \geq 40 \text{ hrs} + \$800 + \text{Tips} + \$V_{dog} - 50 \text{ hrs}$$

This makes the minimum she is willing to accept

$$\$800 + \text{Tips} + \$V_{dog} - 10 \text{ hrs}$$

If Rachel likes having the dog with her, $\$V_{dog} > 0$, so the minimum amount the bookstore would need to pay her goes up.

- Q2)** Barbie, Skipper and Alan live together and split their time between two chores — tidying up rooms and cooking meals. Each of them has 5 hours a week to spend on chores.
- Q2a)** [5 points] **Claim:** If all three can cook 2 meals in one hour, they must have the same Opportunity Cost of an extra meal. Agree, disagree, or it depends? Explain your reasoning.

Solution: Disagree. Cooking at the same rate does not mean facing the same Opportunity Cost.

Each of them requires half an hour for an extra meal. However, the Opportunity Cost of that meal is the number of tidy rooms given up, and that depends on how many rooms each of them can tidy in half an hour. They will have the same Opportunity Cost only if they also tidy the same number of rooms in an hour.

- Q2b)** [6 points] Right now, as a group, they are efficiently producing 14 meals a week with the rest of their time devoted to tidying rooms. Alan is doing only cooking. Barbie is doing only tidying. **Claim:** If they need an extra meal, Barbie should produce it. Agree, disagree, or it depends? Explain your reasoning.

Solution: It depends — and usually the claim is wrong.

If the group is producing efficiently and Barbie is only tidying, it must be that she has the *highest* Opportunity Cost of meal production. She should not be the one switching if Skipper still has time split between tidying and cooking: Skipper has a lower Opportunity Cost and should be switched in first.

The only reason to switch Barbie in is if Skipper is already cooking full time and has no more time to move into cooking. Since the question doesn't tell us how Skipper's time is split, we can't rule that out.

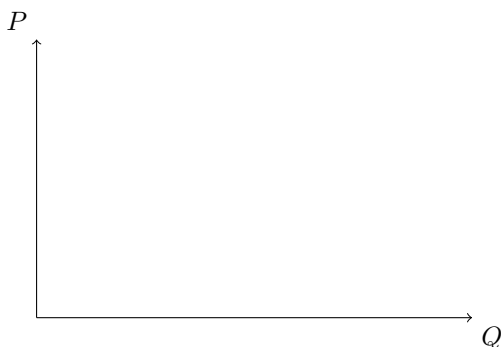
- Q2c)** [6 points] Ken lives by himself and has to split his time between the same chores. Ken's Opportunity Cost of 1 tidy room is 3 meals cooked. He participates in the neighbourhood chore exchange group. At the posted exchange rate of 1 tidy room for 4 meals, neighbours are encouraged to trade chores. Will participating in this group at the posted exchange rate make Ken better off? Explain your reasoning.

Solution: Yes, as long as he trades in the right direction: he should sell tidy rooms, not buy them.

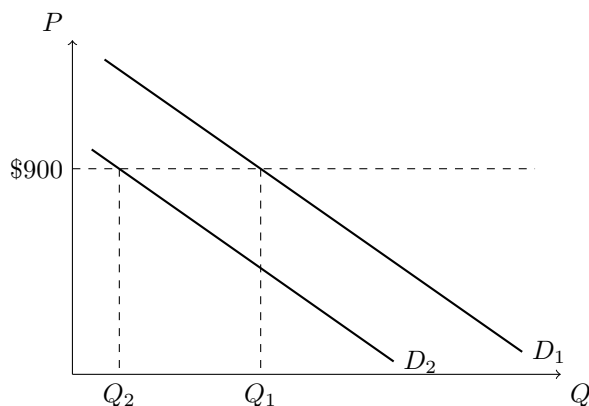
If Ken *buys* tidy rooms, the most he is willing to pay is his own Opportunity Cost of producing one, which is 3 meals. The posted price of 4 meals is too high, so buying would make him worse off. But if he *sells* tidy rooms, each one costs him 3 meals to produce and he gets 4 meals in return. He should sell tidy rooms at the given exchange rate: that way he gets 1 extra meal for the same number of tidy rooms consumed, which makes him better off.

Q3) Ed Sheeran *and* Taylor Swift are holding a special joint concert in Toronto on Nov 30. Tickets to the concert were sold out last year. However, it is still possible to buy tickets on the ticket resale website SeatSwap. Sellers are fans who bought tickets last year at the posted price of \$250 a ticket (non-refundable). They had planned on going to the concert, but could be incentivized to sell their ticket for a high enough price. Market demand is downward-sloping and market supply is upward-sloping in the ticket resale market. The current price of tickets in the resale market is \$900.

Q3a) [5 points] On Oct 20 it is announced that Taylor Swift has temporarily lost her voice due to a cold and will not perform for three months. This means only Ed Sheeran will do the concert on Nov 30. What is your prediction for market quantity demanded for tickets to this concert in the resale market at the same price of \$900 after this announcement? Why? Explain your reasoning. If needed, you can use a diagram to help you think through the question; however, a diagram is not an answer. Your explanation should be sufficient without a diagram.



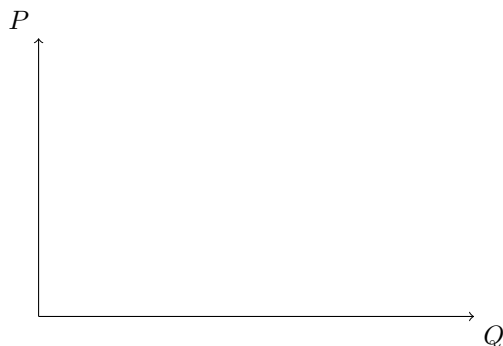
Solution: Quantity demanded falls at the same price of \$900.



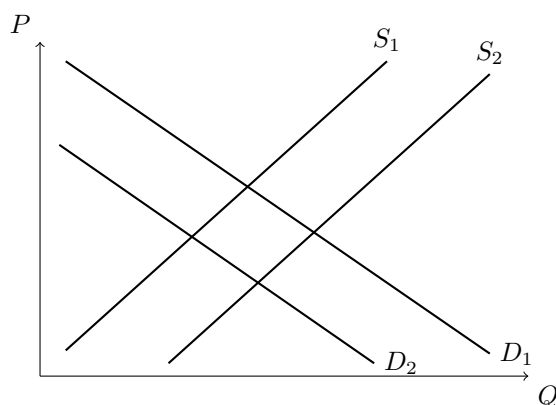
Attending the concert would have given consumers the chance to see two pop superstars; now it gives them only one. If this reduces the value of the concert, it reduces the MWTP at every quantity and demand shifts in. At the same price, fewer units pass the MWTP-versus-price threshold, so quantity demanded decreases at the same price of \$900.

Q3b) [8 points] Suppose that after the announcement, ticket prices in the resale market have decreased. **Claim:** Equilibrium market quantity in the resale market must also decrease. Agree, disagree, or it depends? Explain

your reasoning. If needed, you can use a diagram to help you think through the question; however, a diagram is not an answer. Your explanation should be sufficient without a diagram.



Solution: Disagree. The price definitely falls, but market quantity could go either way.



Demand decreases as MWTP decreases, and that does move us down along the supply curve — but the entire supply curve moves as well. People holding tickets may be more willing to sell them: they are giving up the chance to see a less valuable concert, and so are willing to accept a lower amount at every quantity. Supply moves out.

A decrease in demand and an increase in supply both push the market price down, which is consistent with the observed price fall. But they push quantity in opposite directions, so the net effect on market quantity is uncertain — it depends on which curve moves relatively more. Quantity need not decrease.

Q3c) [5 points] Consider the perfectly competitive market for Taylor Swift (TS) hoodies in Toronto. Demand is downward-sloping and supply is upward-sloping. Suppose fans attending a concert to see her usually buy TS hoodies to wear specifically for the concert. When it is announced that TS will not be doing the concert, just Ed Sheeran is, market price and consumer spending on TS hoodies fall. **Claim:** As consumer spending decreases when price decreases, market demand for TS hoodies must be inelastic. Agree, disagree, or it depends? Explain your reasoning.

Solution: Disagree. The elasticity of demand is irrelevant here.

A decrease in MWTP moves the demand curve in, and we go to a lower price *and* a lower quantity along the supply curve. A decrease in both price and quantity decreases expenditure whatever the elasticity of demand. The elasticity would matter if we were moving *along* a fixed demand curve, but here the curve itself has shifted.